

COMP 1110	
Term	Fall 2024
Course Name	Introduction to Computer Programming
Assignment	3
Due Date	November 9, 2024 @23:59
Marks	30 (zero mark for late submission)
Submission Instructions	1. Your program should be well-written 2. Submit Jupiter notebook file or .py file and name it as FirstNameAssignment-3 <i>Additional instructions:</i> 1. Add proper comments to your code to enhance readability and explain what the code is doing 2. Use the Camel case naming convention for naming the variable names.

Q1: Given a list containing a range of numbers from **0 to n** with a missing number, find the missing number in the input list. (6 points – function definition (1 point), use of range function (1 point), use of loop (2 points), docstring and comments (1 point), correct output (1 point))

Eg: If your input list is: [1,2,4,6,7,8,10]
The output list should be: [3,5,9]

Q2: Write a Python program with a function that takes two parameters to split a given list into two parts where the length of the first part of the list is taken from

the user. (6 points – function definition with two parameters (1 point), user input (1 point), list slicing (3 points), comments and docstring (1 point))

Eg: Input: [3,5,6,8,9,2,6,1,8]

Size of first part = 4

Output: [3, 5, 6, 8], [9, 2, 6, 1, 8]

Q3: Write a function that takes one parameter to remove duplicate words from a given list of numbers.(6 points – function definition with one parameter(2 points), use of loop (2 points), use of built-in functions (1 point), docstring and comments (1 point))

Eg: input -- [1, 4, 5, 7, 8, 4, 4, 6,7,1]

Output -- [1, 4, 5, 7, 8, 6]

Q4: Write a Python function to create and print a list where the values are the squares of numbers between 1 and 30 (both included). (4 points – correct function definition (2 points)s Use of range function (1 point), docstring and comments (1 point)).

Q5: Dr. Riya gives a series of exams during the semester in her COMP1110 class. At the end of the term, she drops each student's lowest test score before averaging the scores. She has asked you to design a program that will read a student's test scores as input and calculate the average with the lowest score dropped. Use the given algorithm to write your program. (8 Points – comments and docstring (2 points))

Step 1: Get the student's test scores (1 point)

Step 2: Calculate the total of the scores (1 point)

Step 3: Find the lowest score (1 point)

Step 4: Subtract the lowest score from the total, which is our adjusted total (1 point)

Step 5: Divide the adjusted total by one less than the number of test scores, which is our average (1 point)

Step 6: Display the average (1 point)